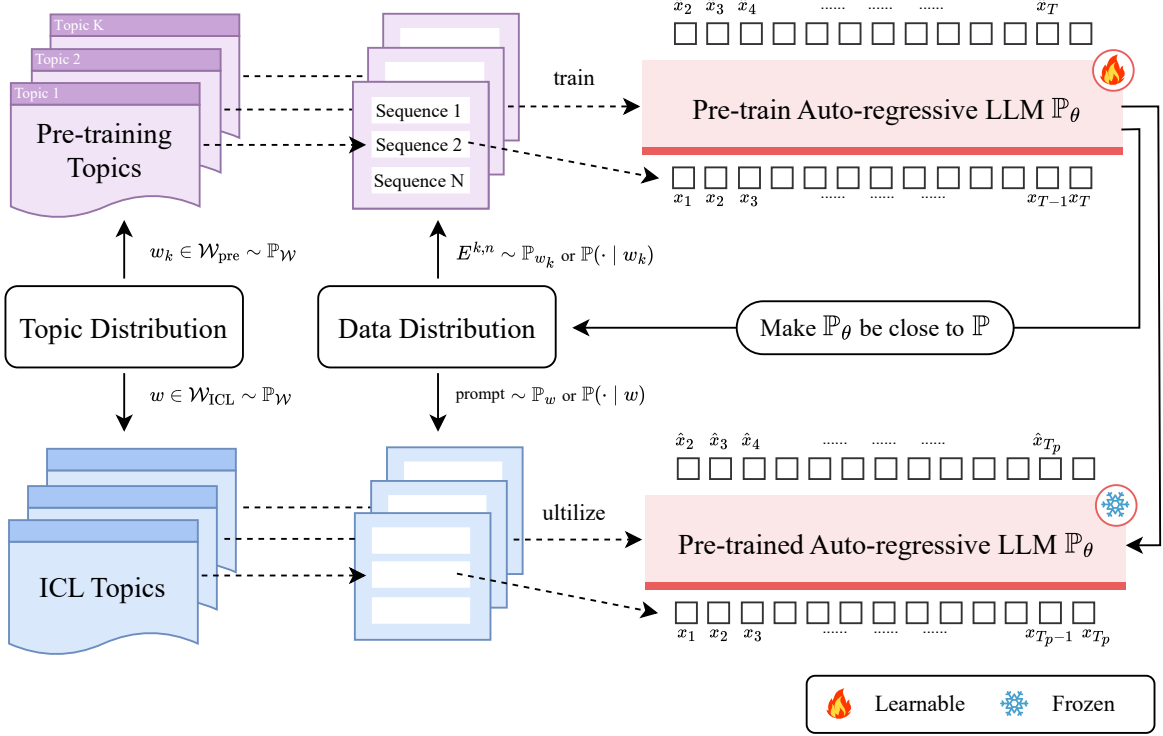
Topic K
Topic 2
Topic 1
Pre-training Topics
Sequence 1
Sequence 2
Sequence N
train
Pre-train Auto-regressive LLM $\mathbb{P}_\theta$
$\hat{x}_2$ $\hat{x}_3$ $\hat{x}_4$ .... $\hat{x}_T$
x_1 x_2 x_3 .... x_{T-1} x_T
$w_k \in \mathcal{W}_{\text{pre}} \sim \mathbb{P}_{\mathcal{W}}$
$E^{k,n} \sim \mathbb{P}_{w_k}$ or $\mathbb{P}(\cdot \mid w_k)$
Topic Distribution
Data Distribution
Make $\mathbb{P}_\theta$ be close to $\mathbb{P}$
$w \in \mathcal{W}_{\text{ICL}} \sim \mathbb{P}_{\mathcal{W}}$
prompt $\sim \mathbb{P}_w$ or $\mathbb{P}(\cdot \mid w)$
ICL Topics
ultilize
Pre-trained Auto-regressive LLM $\mathbb{P}_\theta$
$\hat{x}_2$ $\hat{x}_3$ $\hat{x}_4$ .... $\hat{x}_{T_p}$
x_1 x_2 x_3 .... x_{T_p-1} x_{T_p}
Learnable
Frozen